

Repair Manuals

Timing chain: removal/installation

Warnings and recommendations

Unless otherwise advised by the manufacturer, the following procedures are recommended:

Always renew the timing chain

If the timing chain is to be re-used, mark the running direction before removal

Never turn the engine in the direction opposite to that of normal operation

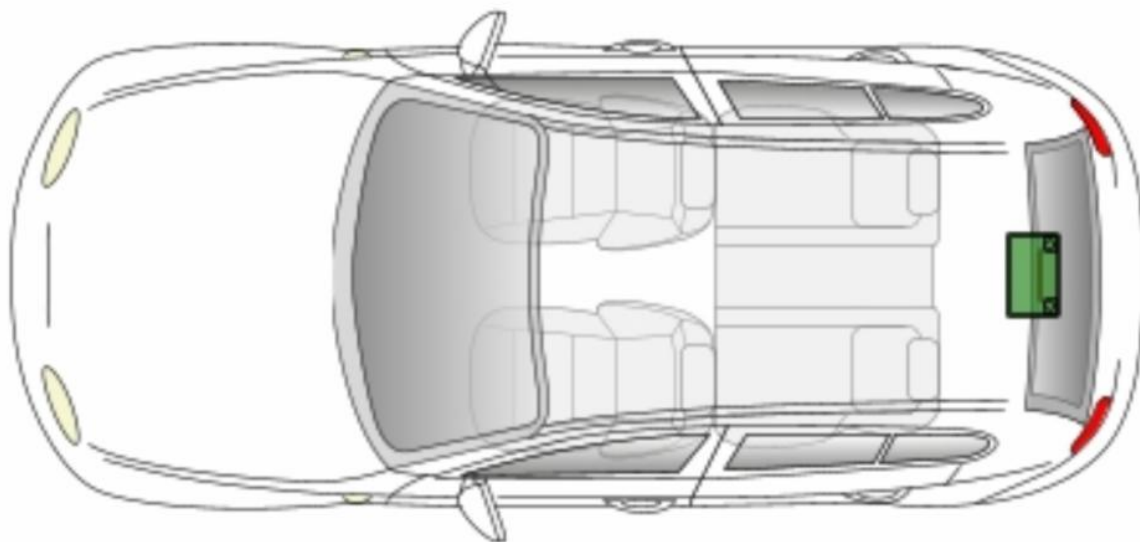
Note: Avoid bending, torsion or traction

Always use new gaskets

General

Before disconnecting the battery cable, check the audio system security code

Battery location



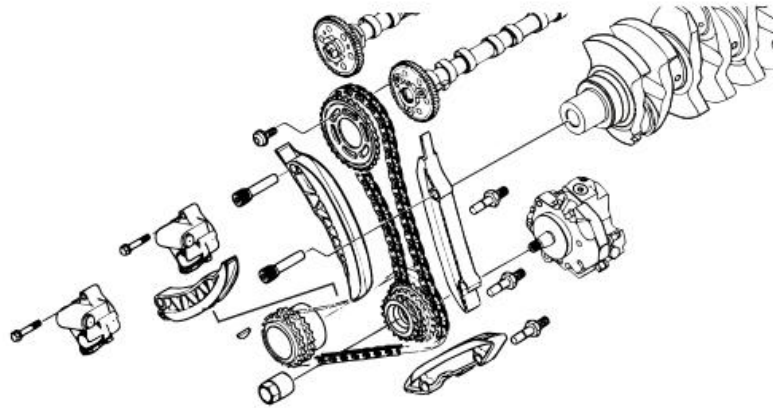
Luggage compartment

Disconnect the battery earth cable

Apply sealant

Renew the seals

Repair Manuals



Check that the crankshaft locking tool can be fitted

5-speed gearbox

Remove the gearbox

Removal

Use a hoist to support the engine

Lower the front axle subframe

Drain the engine oil

Sump to gearbox:

Remove the bolts

Oil sump to engine:

Remove the remaining bolt(s)

Remove the oil sump

Remove the cylinder head

Remove the front splash guard

Remove the reinforcement plate

Remove the fan cowling

Remove the air-conditioning belt

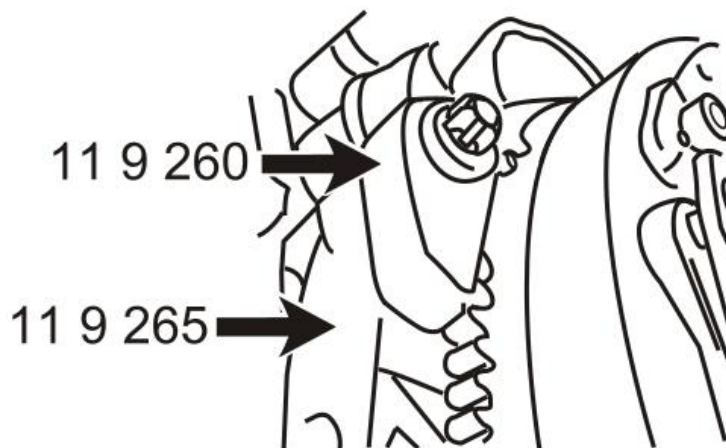
Remove the alternator belt

Remove the power steering pump belt

Disconnect the gearbox exhaust support

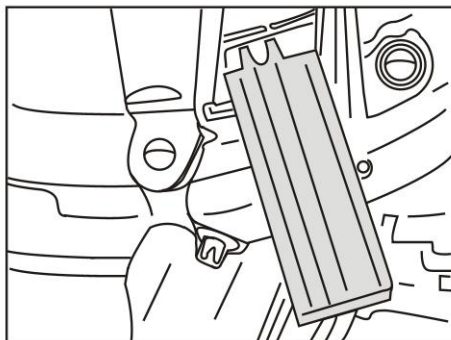
Vehicles with manual gearbox:

Fit the crankshaft locking tool (11 9 260, 11 9 265)



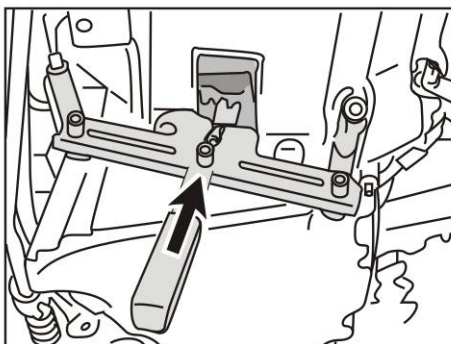
Repair Manuals

Fit the crankshaft locking tool (11 8 010)



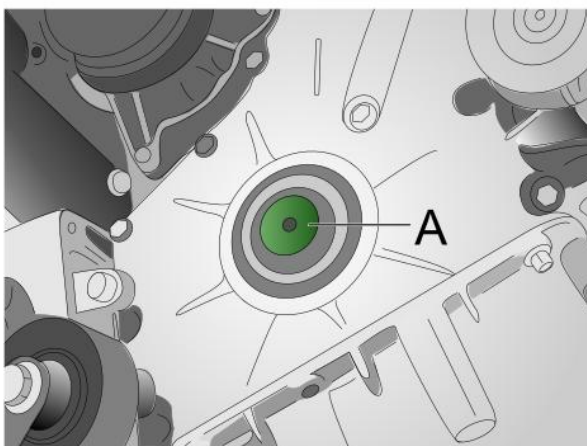
Vehicles with automatic transmission (6HP-19)

Fit the crankshaft locking tool (11 8 183, 11 8 182, 11 8 185)



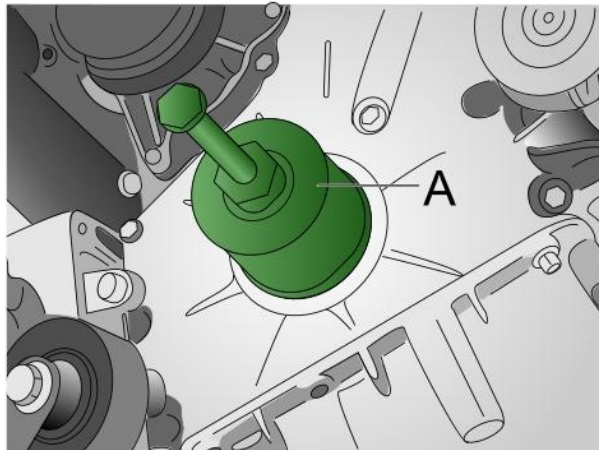
Remove the crankshaft pulley

Fit the special tool (A) (11 1 210, 11 1 212)



Remove and discard the crankshaft oil seal

Use a special tool: (A) (11 1 210, 11 1 212)

Repair Manuals

Remove the power steering pump pulley

Remove the power steering pump

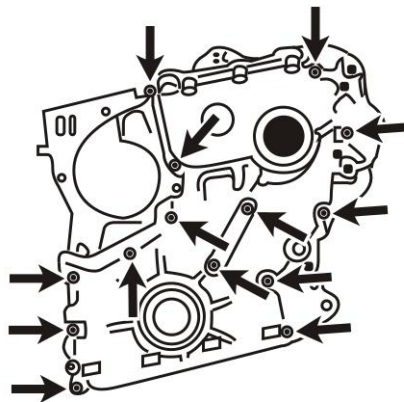
Note: Do not disconnect the lines

Remove the ancillary drive belt tensioner

Remove the coolant pump

Remove the alternator

Remove the timing chain cover



Lock the chain tensioner with a locking pin (1) (11 3 340)

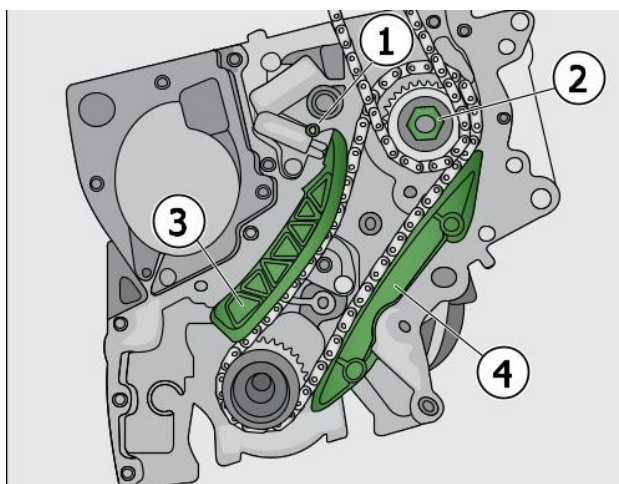
Remove the chain tensioner rail(s) (3)

Remove the guide rail(s) (4)

Remove the high-pressure pump nut (2)

Remove the timing chain and gearwheels together

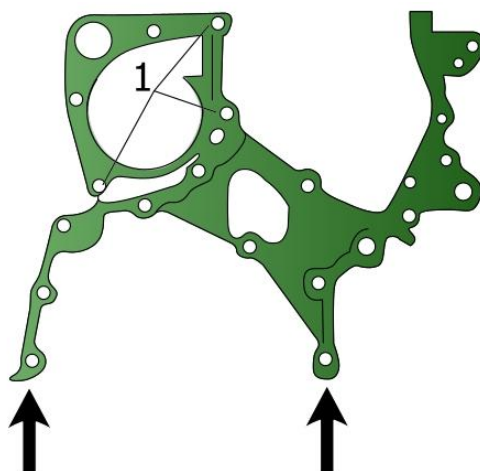
Repair Manuals



Installation

Check the dowel pin(s) for damage

Cut the gasket at the marked points (1)



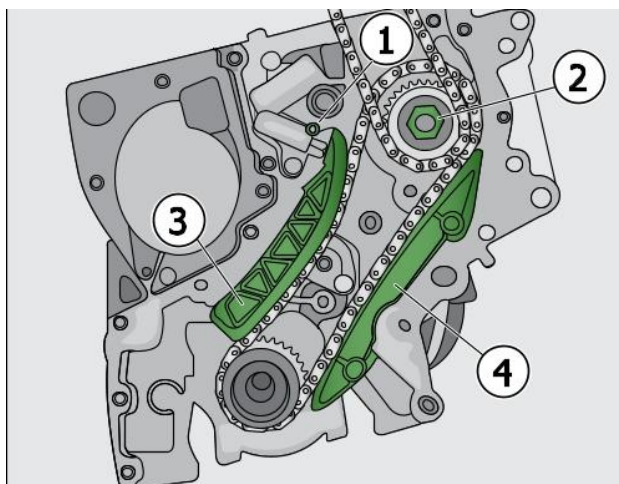
Fit the timing chain and gearwheels together

Refit the guide rail(s) (4)

Refit the chain tensioner rail(s) (3)

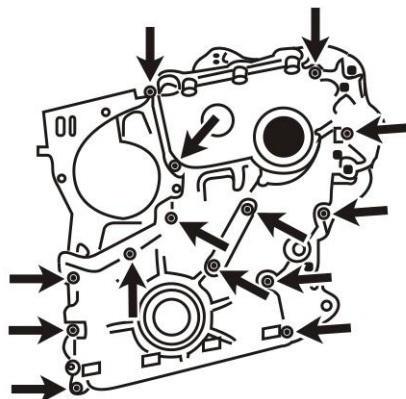
Refit the nut (2: 65 Nm)

Remove the locking pin from the chain tensioner (1 = 11 3 340)

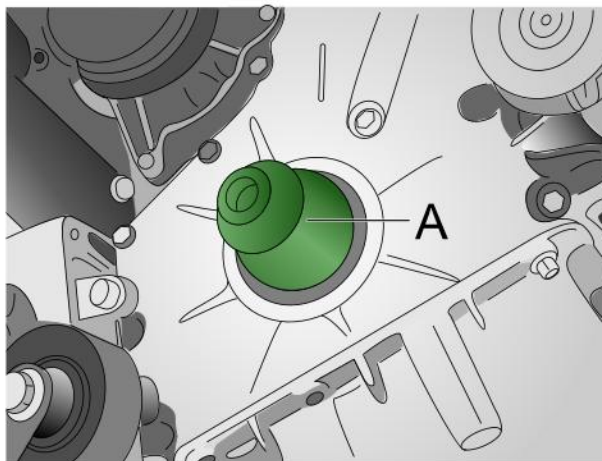


Repair Manuals

Refit the timing chain cover (M6: 10 Nm, M7: 15 Nm, M8: 22 Nm, M10: 47 Nm)



Fit the special tool: (A = 11 7 400)

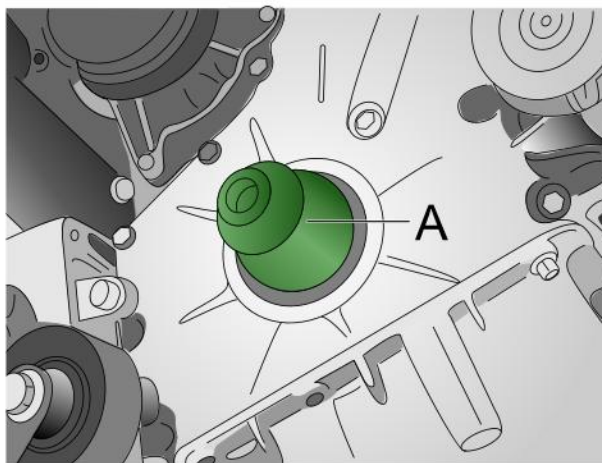


Apply grease

Fit the crankshaft seal over the special tool

Note: Do not touch the lip of the seal

Remove the special tool (A = 11 7 400)



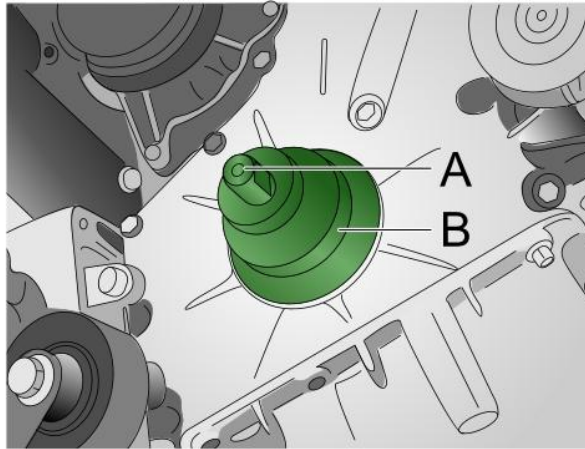
Fit the special tool: (B = 11 5 110) (11 5 110)

Tighten the bolt (A = 11 7 410)

Note: Until the crankshaft seal is level with the timing cover

Repair Manuals

Remove the special tools (11 7 410, 11 5 110) (11 5 110)



Refit the alternator

Refit the coolant pump

Refit the ancillary drive belt tensioner

Refit the power steering pump

Refit the power steering pump pulley

Refit the crankshaft pulley

Use new bolt(s) (M18 x1.5)

Stage 1: (100 Nm)

Stage 2: (150°)

Remove the crankshaft locking tool

Reconnect the gearbox exhaust support

Refit the power steering pump belt

Refit the alternator belt

Refit the air-conditioning belt

Refit the fan cowl

Refit the reinforcement plate

Refit the front splash guard

Refit the cylinder head

Refit the oil sump (M6 8.8: 10 Nm, M6 10.9: 12 Nm, M8 8.8: 22 Nm)

Lift the front axle subframe

Remove the engine support tool

Refill the engine oil

Engine sump, including filter 5.5 (l)

Reconnect the battery earth cable

Torque settings

High-pressure pump gearwheel

High-pressure pump gearwheel 65 (Nm)

Timing cover

Timing cover (M6) 10 (Nm)

Timing cover (M7) 15 (Nm)

Timing cover (M8) 22 (Nm)

Timing cover (M10) 47 (Nm)

Oil sump

Sump (M6 - 8.8) 10 (Nm)

Sump (M6 - 10.9) 12 (Nm)

Sump (M8 - 8.8) 22 (Nm)

Crankshaft pulley

Crankshaft pulley (Renew the bolt)

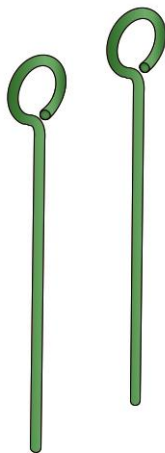
Repair Manuals

Stage 1 100 (Nm)

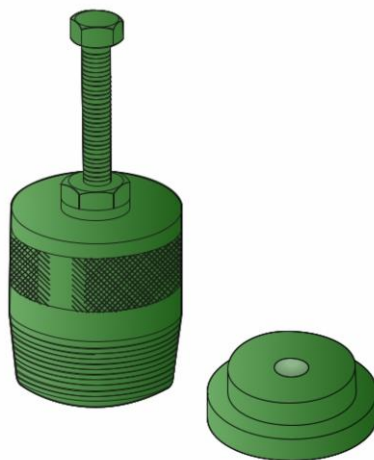
Stage 2 150 (°)

Special tools

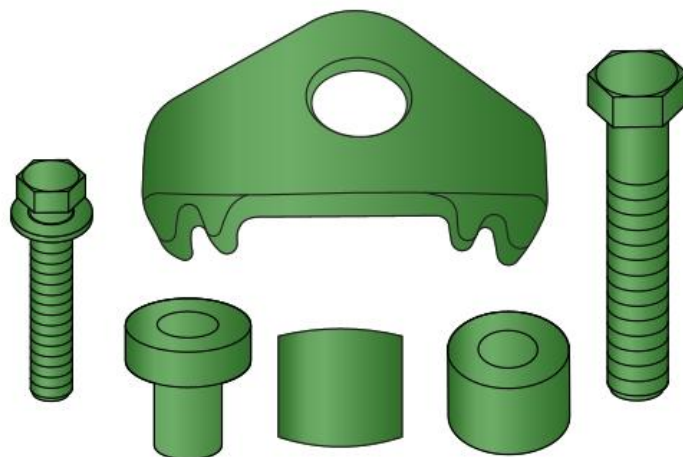
Chain tensioner locking pin: (11 3 340)



Extractor: (11 1 210, 11 1 212)

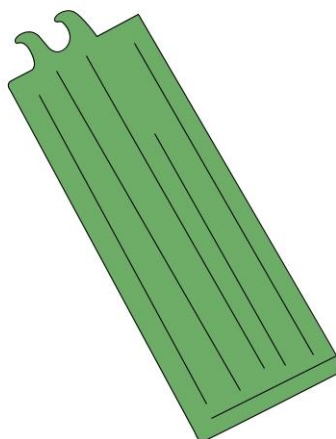


Flywheel locking tool: (11 9 260, 11 9 265)

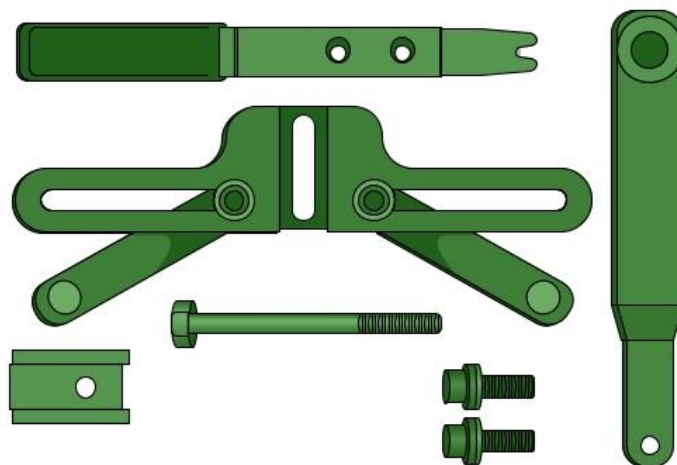


Crankshaft locking tool: (11 8 010)

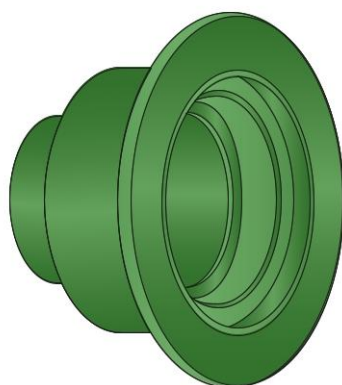
Repair Manuals



Automatic transmission: (11 8 182, 11 8 183, 11 8 185)

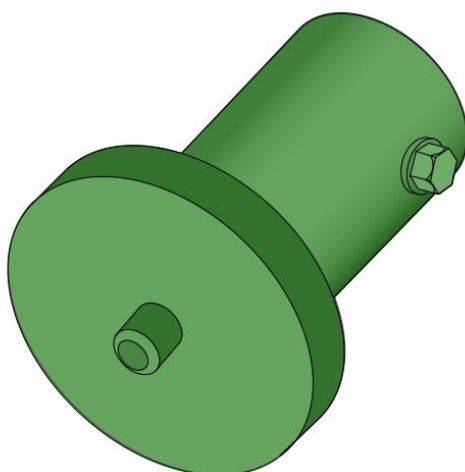


Centre bush: (11 5 110)

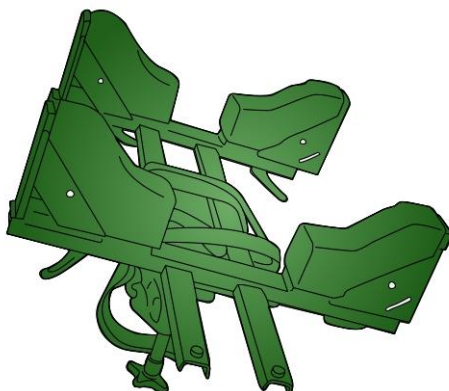


Support (31 2 220)

Repair Manuals

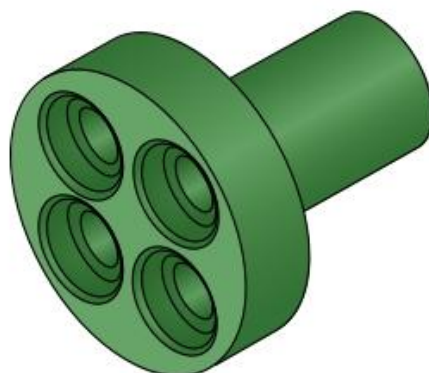


Gearbox support: (23 0 132, 00 2 030, 23 4 050)



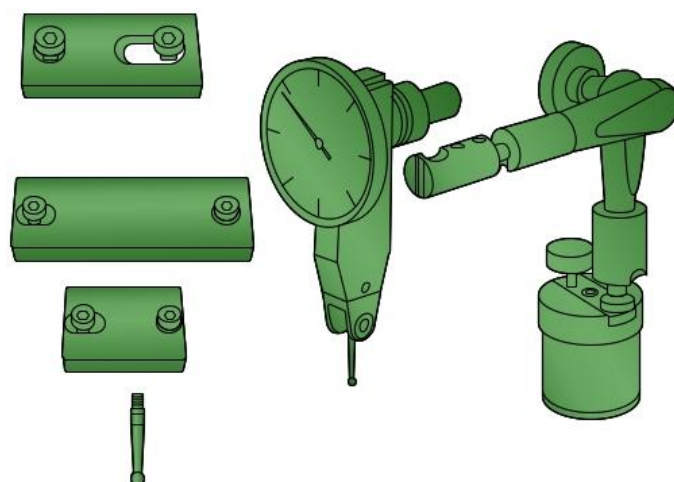
Gearbox support: (23 4 050, 23 0 132, 00 2 030)

Socket: (11 6 480)

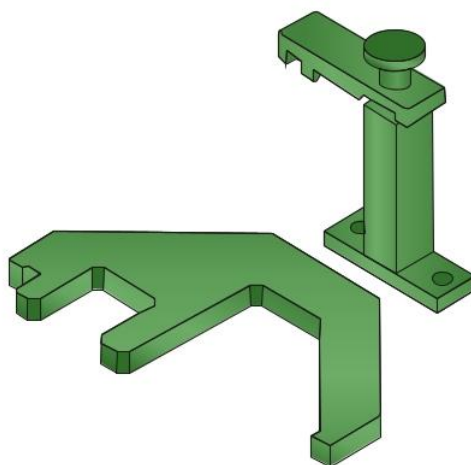


Special tool: (11 6 252, 11 6 254)

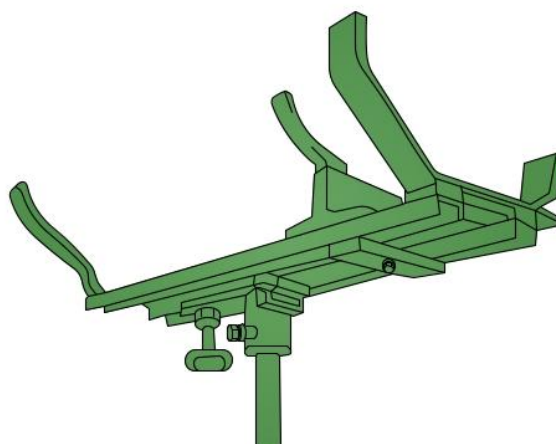
Repair Manuals



Camshaft alignment tool: (11 6 321, 11 6 322)



Gearbox support: (23 4 050, 00 2 030)



Crankshaft locking pin: (11 2 300, 11 5 180)

Repair Manuals

